Spectroscopy and Atomic Physics: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

15 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Spectroscopy and Atoms: worked solutions

Solution 1 — A small redshift. Return →

$\beta = 100/299792.458 = 3.33564 \times 10^{-4}$. The relativistic factor gives **656.49895 nm**, a shift of 0.21895 nm. The low-speed formula gives 656.49891 nm, differing by only 3.7×10^{-5} nm.

Check. Recession shifts the line redward. At this speed the approximation is excellent.

Solution 2 — Two stars in one spectrum. Return →

$v \simeq c\Delta\lambda/\lambda_0$: **−119.9 km/s** and **+179.9 km/s**. Their mean, the systemic velocity, is **+30.0 km/s**. Relative to the system both semiamplitudes are 150 km/s, so $m_1/m_2 = K_2/K_1 = 1$.

Check. The momenta about the barycentre are opposite. At quadrature in the stated circular edge-on orbit, the observed speeds are the orbital semiamplitudes.

Solution 3 — Weighing a rotating galaxy. Return →

Idea. Remove systemic motion, then deproject. $v_{\rm rot} = 200/\sin 30^\circ = \mathbf{400\ km\,s^{-1}}$. For circular motion, $M = v^2 R/G$. With $R = 5$ kpc, $M = 1.86 \times 10^{11} M_\odot$.

Check. The mass is enclosed mass under the spherical-equivalent estimate; a disk geometry changes the coefficient. The two observed sides would lie at about 4800 and 5200 km/s.

Solution 4 — Thermal width. Return →

$\sigma_v = \sqrt{k_B T/m_H} = \mathbf{9.08\ km\,s^{-1}}$. FWHM $= 2.35482\sigma_v = \mathbf{21.39\ km\,s^{-1}}$. In wavelength, $\Delta\lambda = 656.28(21.39/299792.458) = \mathbf{0.04682\ nm}$.

Check. This is a width, not a bulk shift. Heavier atoms at the same temperature have smaller thermal speeds.

Solution 5 — Can the spectrograph separate them? Return →

$\Delta\lambda = \lambda/R = \mathbf{0.025\ nm}$; equivalent velocity width $c/R = \mathbf{15.0\ km\,s^{-1}}$. A 0.040 nm separation is larger than the instrumental FWHM, so the doublet is **resolvable in principle**. Real contrast and line widths still matter.

Check. Higher resolving power means a smaller instrumental width.

Solution 6 — A finite lifetime. Return →

With $\Delta E \sim \hbar/(2\tau)$ and $\Delta E = h\Delta\nu$, the half-width at half-maximum is $\Delta\nu_{\rm HWHM} \sim 1/(4\pi\tau) = \mathbf{7.96\ MHz}$; the corresponding full width is $\mathbf{15.9\ MHz}$. Using the half-width, $\Delta\lambda \simeq \lambda^2\Delta\nu/c = \mathbf{6.64 \times 10^{-6}\ nm}$.

Check. Always state whether a linewidth is HWHM or FWHM. A precise Lorentzian width depends on whether the quoted lifetime refers to amplitude or population decay; the key physics is that shorter lifetime means broader line.

Solution 7 — A magnetic fingerprint. Return →

$\Delta\nu = eB/(4\pi m_e) = \mathbf{1.400\ GHz}$ for each shifted component. Near 500 nm, $\Delta\lambda = \lambda^2\Delta\nu/c = \mathbf{0.001167\ nm}$ (0.01167 Å). The two outer components are separated by **2.800 GHz**, or **0.002334 nm**.

Check. The unshifted component remains at the centre in the normal triplet. Real atomic structure can produce anomalous splitting.

Solution 8 — Two polarizers. Return →

For already polarized light, $\cos^2 60^\circ = \mathbf{0.25}$. For unpolarized light, the first polarizer leaves 0.5 and the second leaves another 0.25: **0.125** of the original intensity.

Check. The first ideal polarizer both selects a direction and removes half the unpolarized power.

Solution 9 — Brewster's angle. Return →

$\theta_B = \tan^{-1}(1.5/1) = \mathbf{56.3}^\circ$ from the normal. In the ideal dielectric case the reflected beam is fully linearly polarized perpendicular to the plane of incidence.

Check. The reflected and refracted rays are perpendicular at Brewster's angle. Absorption, roughness and multiple surfaces reduce the ideal purity.

Solution 10 — More atoms, not always twice the depth. Return →

Optically thin: equivalent width $W \propto N$. Saturated: the centre is already dark, so W grows slowly as the line widens. Damping wings: strong Lorentzian wings eventually give approximately $W \propto \sqrt{N}$ for fixed damping. Depth alone misses width and saturation, and also depends on instrumental resolution.

Check. Equivalent width is an area; two lines with the same depth can have very different strengths.

Solution 11 — How transparent? Return →

$e^{-0.2} = \mathbf{0.819}$ and $e^{-2} = \mathbf{0.135}$. Half transmission requires $\tau = \ln 2 = \mathbf{0.693}$.

Check. Optical depth adds through successive screens; transmitted fractions multiply.

Solution 12 — Populating an excited level. Return →

With equal weights, $n_2/n_1 = e^{-2/(k_B T)}$. At 6000 K, $k_B T = 0.5170$ eV and the ratio is **0.0209**. At 12,000 K, $k_B T = 1.0341$ eV and the ratio is **0.1446**.

Check. Higher temperature populates the excited level more strongly, but it still need not contain most atoms.

Solution 13 — Why blue scatters more. Return →

The ratio is $(800/400)^4 = \mathbf{16}$. In ideal single Rayleigh scattering, polarization is strongest near **90°** scattering angle.

Check. Multiple scattering, large particles and reflected light reduce the simple polarization and colour rules.

Solution 14 — Expansion parallax. Return →

Idea. Proper motion links transverse speed and distance: $v_t = 4.74047\,\mu(''/\mathrm{yr})\,d(\mathrm{pc})$. The observed angular expansion rate is $\mu = 0.10''$/yr. Then $d = 100/(4.74047 \times 0.1) = \mathbf{211\ pc}$. The radius grows by **1 arcsecond**, doubling to 2 arcseconds.

Check. The calculation assumes spherical expansion so that the rim's transverse speed equals the spectroscopic speed. A non-spherical shell breaks that equality. The distance follows from comparing angular and physical expansion rates, not from the angular size alone.

Solution 15 — An unresolved line through a spectrograph. Return →

Idea. A Gaussian's area is depth times $\sigma\sqrt{2\pi}$. Here $\sigma = 0.10/2.35482 = 0.04247$ nm, so $W = 0.4(0.04247)\sqrt{2\pi} = \mathbf{0.04258\ nm}$. Convolution with an equal Gaussian gives FWHM $\sqrt{0.10^2 + 0.10^2} = \mathbf{0.1414\ nm}$. Conservation of area makes the new depth $0.4/\sqrt{2} = \mathbf{0.2828}$.

Check. The instrument makes the line wider and shallower while preserving its equivalent width under the stated ideal conditions. This is why depth alone is resolution-dependent.